



Product Development of a Substrate Mixer and Portioning Device

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Bachelor's thesis
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Substrate for mushroom cultivation

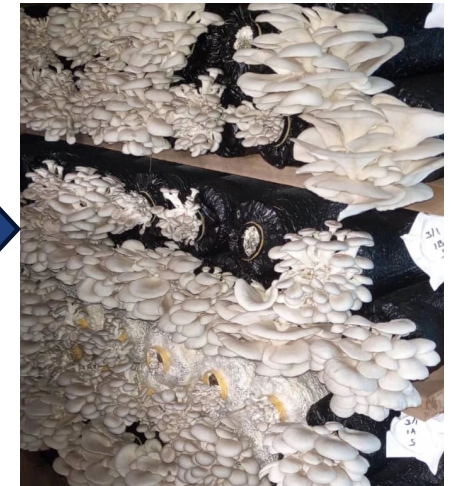
Mix and water agricultural waste products



Bag, pasteurize and insert mushroom spawn



Harvest edible mushrooms



Benefits of mushrooms cultivation in developing countries

Circular Economy: "From waste to money"

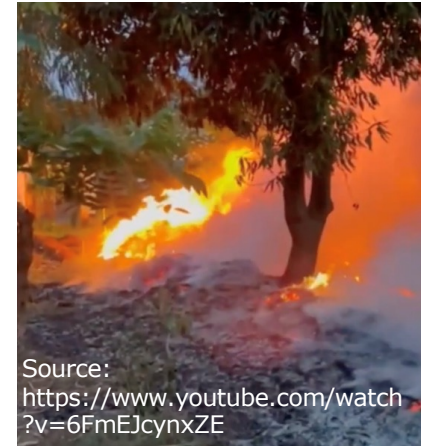
Agricultural waste



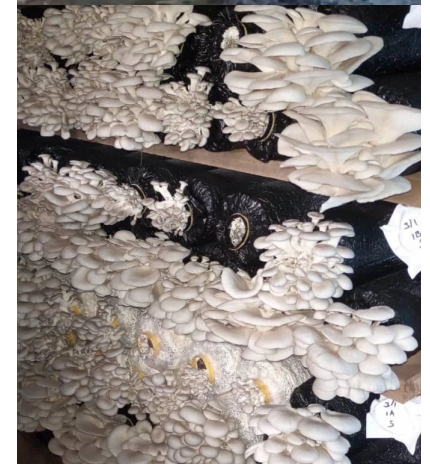
Waste disposal
Environmental pollution

Mushroom growing

Income generating



Source:
<https://www.youtube.com/watch?v=6FmEJcynxZE>



Mushroom cultivation in resource-limited settings

Ugandan baseline



Manual substrate preparation:

- High labor cost
- Inconsistent yields

Mechanized substrate preparation:

- High capital requirement
- High energy consumption

Industrial processing



Mushroom cultivation in resource-constraint settings

Ugandan baseline



Constraints in Kampala:

- Low capital (smallholders)
- Limited material availability
- Unreliable energy supply
- High energy cost

Goal of this project

Industrial processing



Requirements for the device (Research Questions)

Mixing

How can heterogeneous agricultural residues be processed to achieve a homogeneous substrate composition throughout the mixing vessel?

Watering

How can the substrate be uniformly hydrated to achieve a target moisture content of 60-70 wt%?

Portioning

How can the processed substrate be portioned into bags with deviations within a tolerance of ± 10 wt%?

Product Development - Methodology

“Simplicity is the Reproducibility.” – Leonardo da Vinci

Globally available materials



Simple manufacturing



Low-tech & low-cost



Product Development

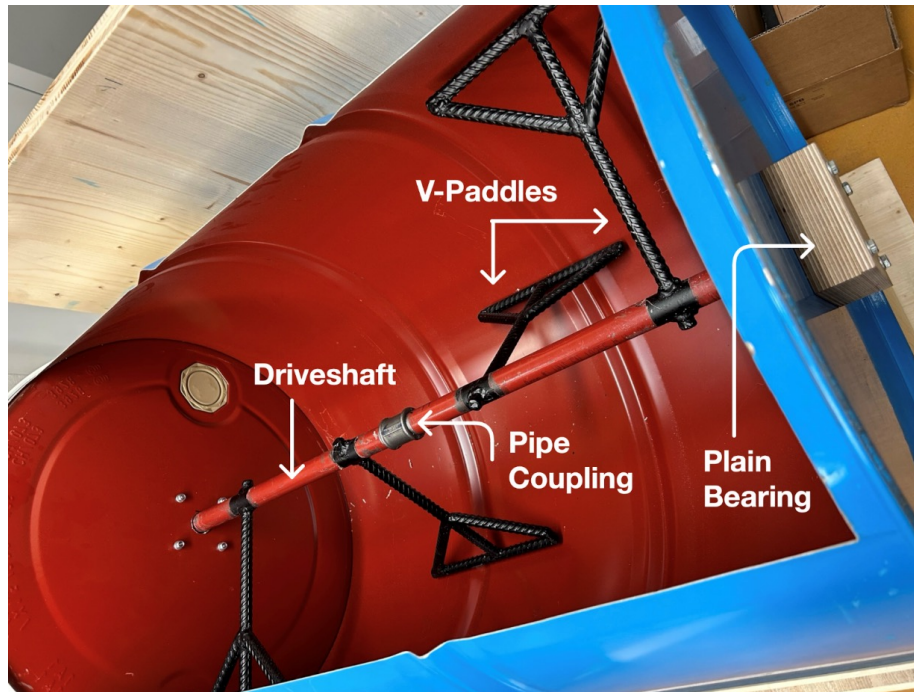


- Oil drum
- Wooden panels
- ✓ Globally standardized materials & items

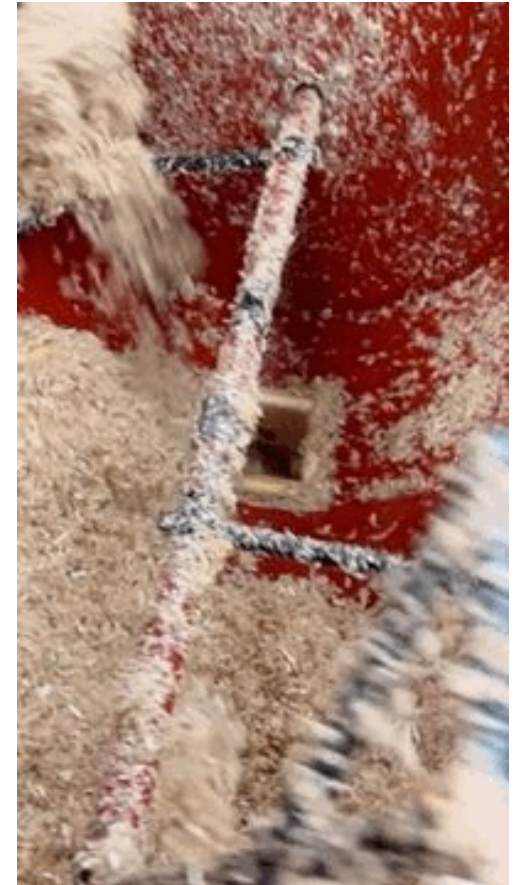
Product Development - Mixing

Final agitator

Model scale 1:5 – Testing different paddle geometries



- Steel pipes
 - Pipe couplings
 - Rebar Paddles
 - Wooden Bearings
- ✓ All materials used on construction sites



Product Development - Watering

- Hydrostatic Reservoir
- Manifold (sprinkler)
- ✓ Gravity driven
(no energy supply)



Product Development - Portioning

Sealing Valve



dosing

Conveyance Duct



bagging

Slide Valve

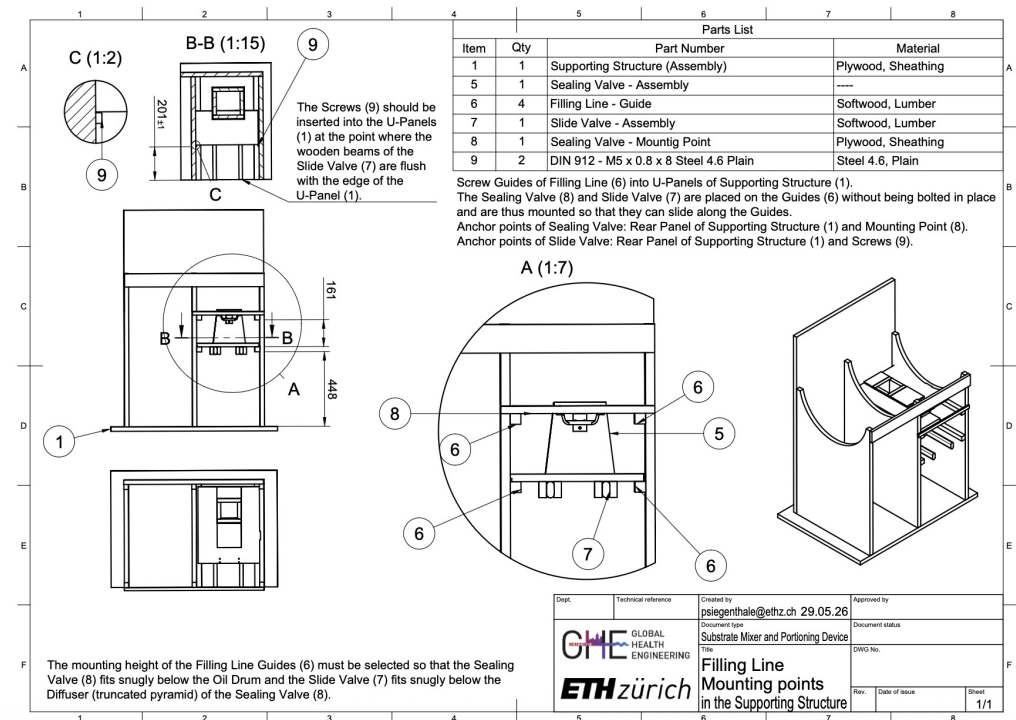
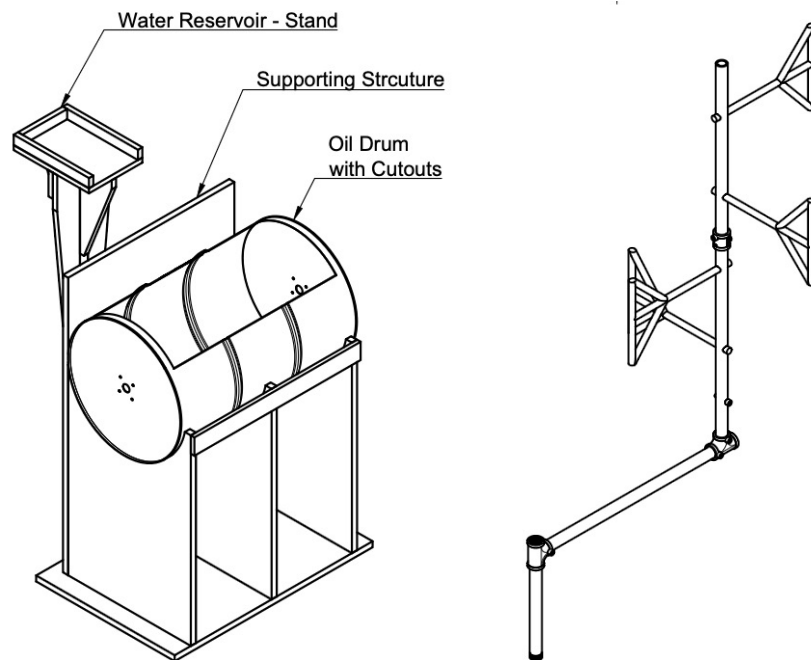


Product Development - Portioning

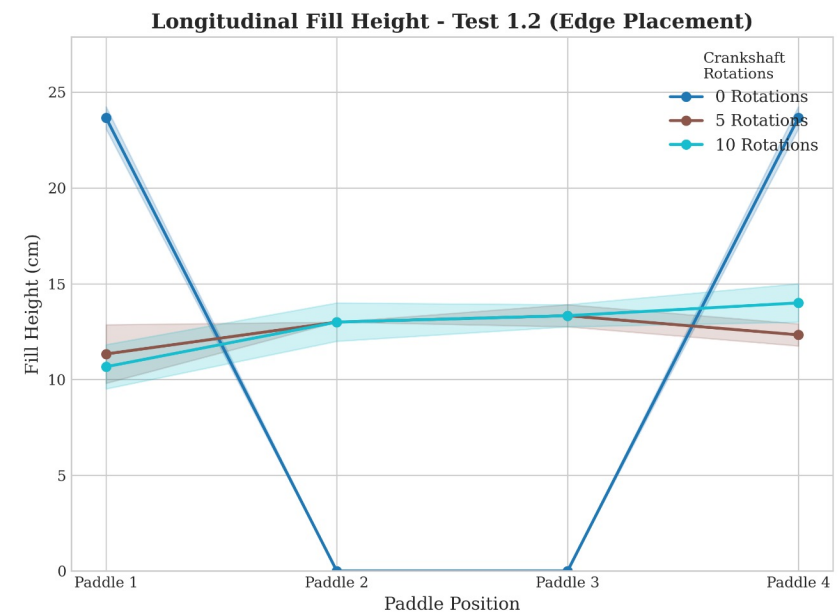
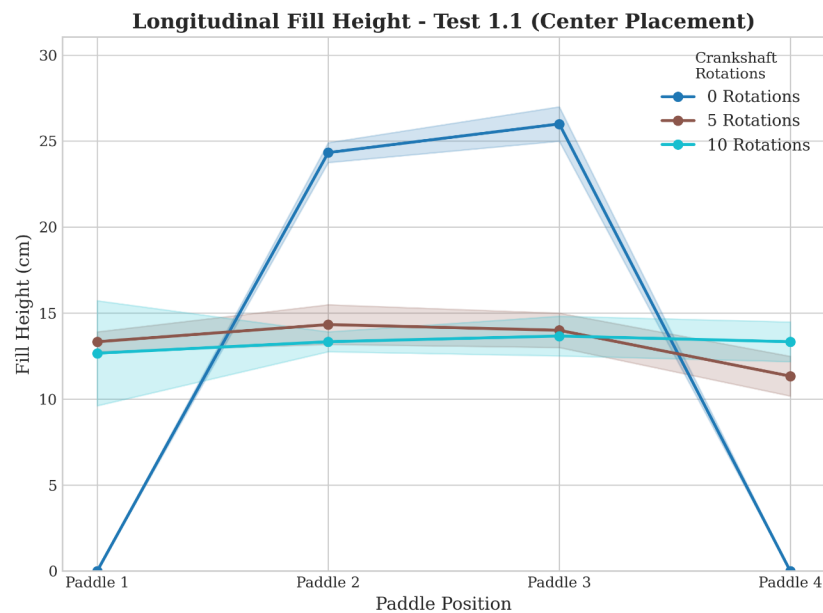
Substrate blockages in conveyance duct



Deliverables – Technical Drawings, User Manual



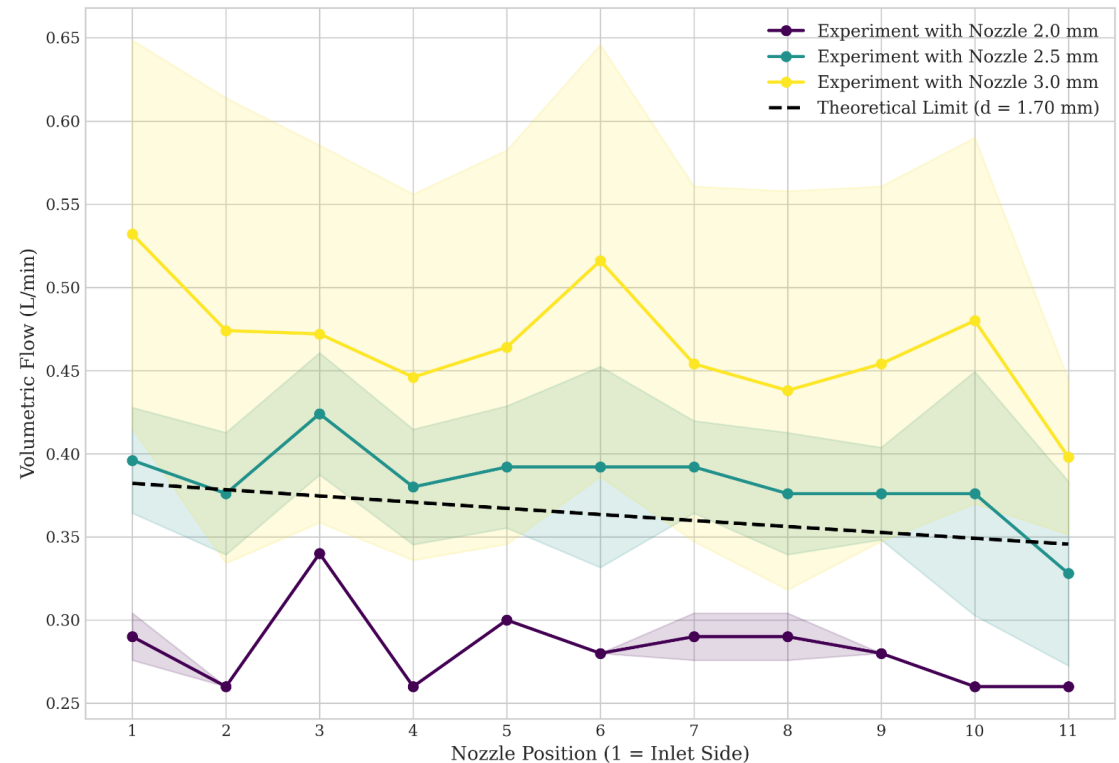
Mixing – Longitudinal Transport Efficiency



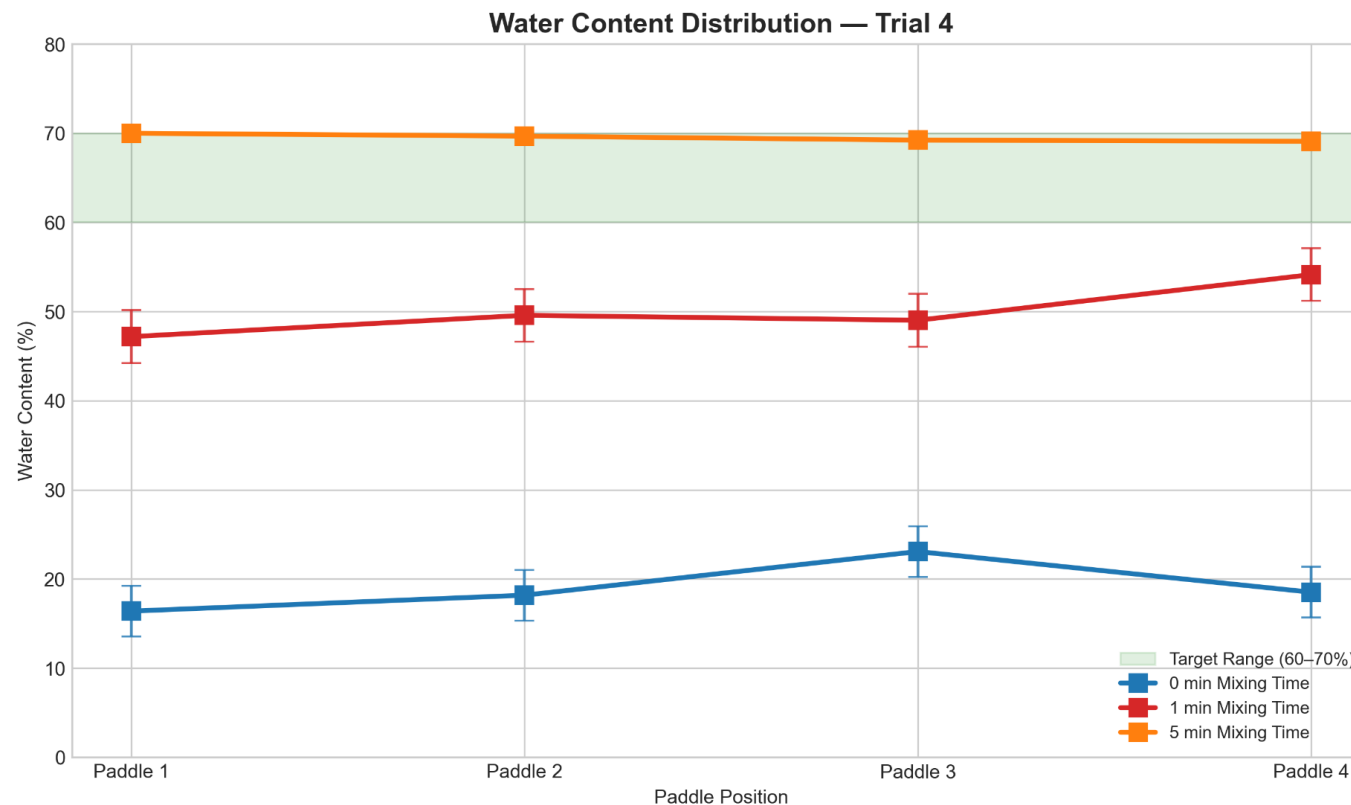
- Equalized fill height already after five agitator rotations.
- Rapid, bidirectional longitudinal transport.

Watering – Nozzle Outflow

- Manufacturing inaccuracies dominate.
- 10% drop threshold not reached (3mm bore).
- Total outflow increased:
4.21 L/min (2.5mm bore) to
5.13 L/min (3.0mm bore)



Mixing & Watering – Water Content Distribution



Mixing Vessel

$(69.50 \pm 0.41) \text{ wt\%}$

Portioned Bags

$(69.55 \pm 0.62) \text{ wt\%}$

Samples deviation (5')

$(0.45 \pm 0.06) \text{ wt\%}$

- Water content very homogeneous distributed.
- Water content between 60-70 wt% reached.

Portioning – Bagging Performance

Trial	Prototype	Mean (kg)	SD (kg)	RSD (%)	Loss (wt%)	Note
1	PE-Pipe	0.414	0.153	36.89	45.42	all bags
2	Diffuser	1.367	0.105	7.69	8.74	all bags
3	Diffuser	1.146	0.186	16.25	3.24	all bags
3	Diffuser	1.235	0.105	8.52	—	bags 6–18

- Diffuser can reach $\pm 10\%$ threshold.
- Sensitive to user errors.

Working Prototype



Hand-driven, low-cost mechanical solutions can successfully standardize substrate processing.

Big reduction of labor estimated.

Outlook – Recommendations

- **Field Validation in Uganda:**
 - Analysis of user acceptance, reproducibility, durability.

ETH zürich

Final Prototype

